

Paper Library App

Local Flask · No cloud · No login

Project Asset Document

Stack: Flask · SQLite · PDF.js · arXiv API · HuggingFace Hub

PROJECT SUMMARY

Name Paper Library App

Deliverable Runnable repo: README.md + requirements.txt

Paper Library App is a lightweight, fully local web application for researchers who want to keep their reading list organised without depending on any cloud service. You drag a PDF onto the page and the app automatically identifies the arXiv paper inside it, pulls the title, authors, year, and category from the arXiv API, and saves everything to a local SQLite database. From there you can open any paper in a built-in PDF viewer that lets you select text and leave persistent highlights, group papers with coloured tags, filter and full-text-search your entire library, and discover new work by querying Hugging Face's papers section and importing results with a single click - all without an account, a token, or an internet connection beyond the initial fetch.

STACK

Layer	Technology	Notes
Backend	Python 3.10+ · Flask	HTTP server, routing, API proxy
Database	SQLite (via sqlite3 / SQLAlchemy)	Papers, tags, highlights persist on disk
PDF view	PDF.js (Mozilla)	In-browser viewer with text selection
arXiv	arXiv public API (export.arxiv.org)	Metadata + PDF fetch - no key required
HF Hub	Hugging Face Hub REST API	Paper search - no token required
Frontend	Vanilla JS + CSS custom properties	Drag-drop, search, tag UI

REQUIRED FEATURES

F1 Drag-Drop + arXiv Lookup

- Drop any PDF anywhere on the page
- App extracts the arXiv ID from the PDF (not the filename)
- Fetches title, authors, year, category, source from arXiv API
- Adds a new row to the library; shows loading indicator during fetch
- Renamed files (e.g. paper.pdf) must still resolve correctly

F2 Built-in PDF Viewer + Highlights

- Clicking PDF button opens the paper inline - no new tab, no download
- User can select and copy text from the viewer
- User can highlight passages with a translucent colour overlay
- Highlights persist across full browser reloads (stored server-side)

F3 Tag System

- Left sidebar lists tags as coloured dots + label
- Six default soft/desaturated colours: Red, Orange, Yellow, Green, Blue, Purple
- Add or remove tags on any paper entirely through the UI
- Clicking a tag filters the library; 'All' entry clears the filter
- Each row shows its assigned tags as coloured chips

F4 Full-Text Search

- Search box near the top; placeholder text 'Search library'
- Filter icon on the right + visible / keyboard-shortcut hint
- Filters by title, authors AND extracted body text from PDFs
- Live filtering as the user types
- Pressing / from anywhere focuses the search box

F5 Hugging Face Paper Import

- Discoverable UI to search HuggingFace papers (button / dedicated view)
- Returns live titles from the HF Hub API - no mock data
- Works with no HF token and no arXiv credentials
- One-click import: fetches PDF + metadata via the associated arXiv ID
- Imported row is identical in shape to a drag-dropped paper

UI LAYOUT

The page is divided into three fixed regions:

Region	Position	Contents
Tags Sidebar	Left column (~180 px)	'Tags' heading, coloured-dot list, filter state
Search Bar	Top of centre column	Input with placeholder, filter icon, / hint
Library List	Centre column (remainder)	Paper rows: checkbox, title, authors, source line, PDF button, tags

Each library row contains: left-edge selection checkbox · bold title (link style) · authors line · arXiv [category] · year · Preprint source line · right-aligned PDF button · coloured tag chips.

COLOUR PALETTE

UI variables (CSS custom properties) and the six default tag colours:

Variable / Name	Hex	Swatch	Use
--accent	#2563EB		Primary blue - links, active states
--text	#0F172A		Near-black body text
--muted	#64748B		Secondary / meta text
--border	#E2E8F0		Card and row borders
--bg	#F8FAFC		Page background
--panel	#FFFFFF		Card / panel background
Tag – Red	#FDA4AF		Soft, desaturated - not neon / primary
Tag – Orange	#FCD34D		Soft, desaturated - not neon / primary
Tag – Yellow	#FDE68A		Soft, desaturated - not neon / primary
Tag – Green	#86EFAC		Soft, desaturated - not neon / primary
Tag – Blue	#93C5FD		Soft, desaturated - not neon / primary
Tag – Purple	#C4B5FD		Soft, desaturated - not neon / primary

PERSISTENCE & DATA MODEL

All user state survives a full server restart. Nothing lives only in process memory or browser localStorage.

Table	Key Columns
<code>papers</code>	arXiv ID, title, authors, year, category, source, extracted text, date added
<code>tags</code>	Tag name, hex colour, display order
<code>paper_tags</code>	Join table: paper_id × tag_id
<code>highlights</code>	paper_id, page number, bounding-box coordinates, colour, created_at

FILE DELIVERABLES

File / Path	Purpose
<code>app.py</code>	Flask app entry-point; all routes & API handlers
<code>requirements.txt</code>	Pinned Python deps for pip install
<code>README.md</code>	Single-command setup + start instructions
<code>data/library.db</code>	SQLite database – papers, tags, highlights
<code>data/pdfs/.pdf</code>	Stored PDF files keyed by arXiv ID
<code>templates/index.html</code>	Main library page (list + sidebar + search)
<code>templates/viewer.html</code>	In-app PDF viewer with highlight support

<code>static/main.js</code>	Frontend JS – drag-drop, search, tag UI
<code>static/style.css</code>	App-wide stylesheet